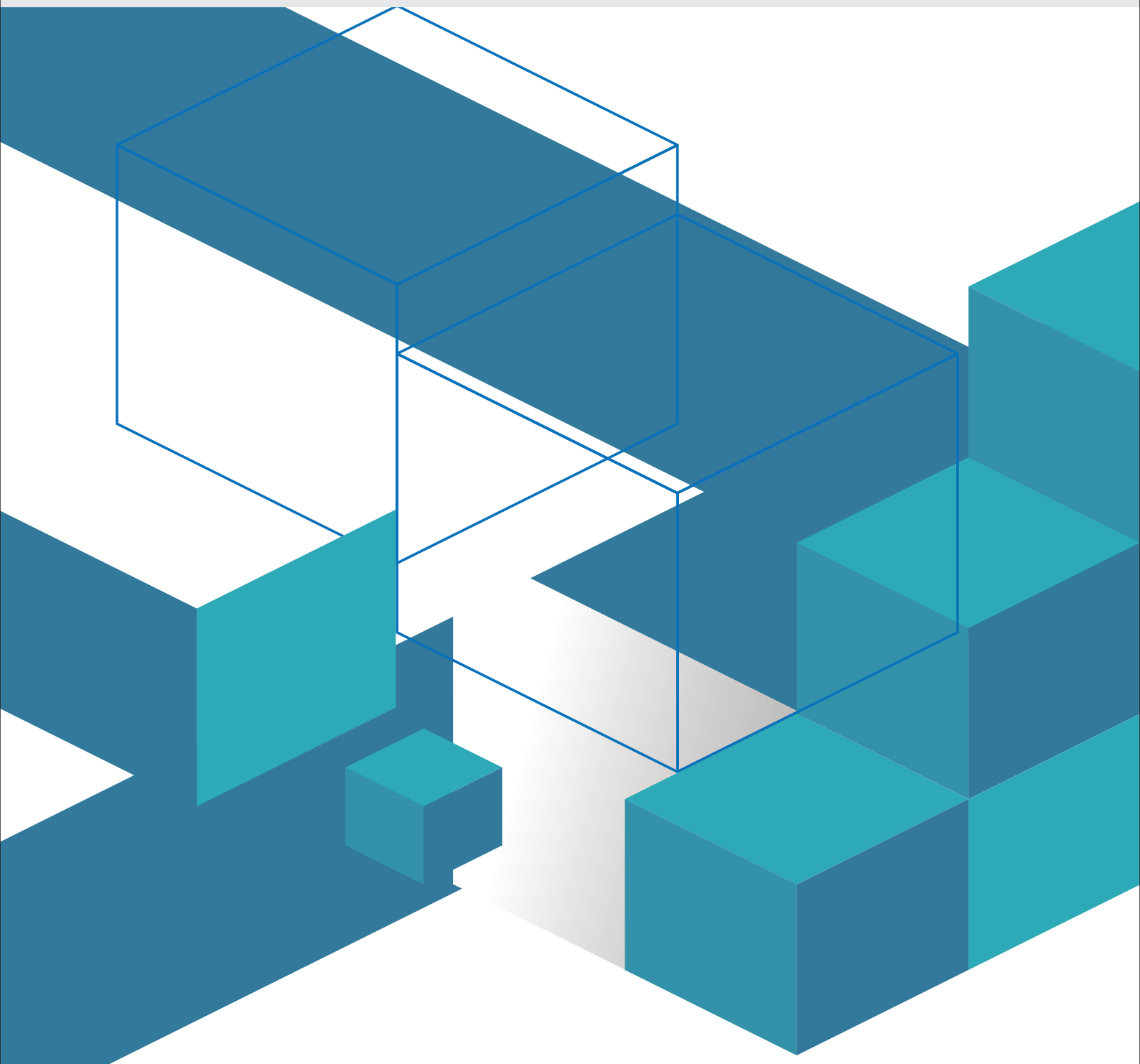
MATHEMATICS

LIMIT, CONTINUITY & DIFFERENTIABILITY_MEGA_XII



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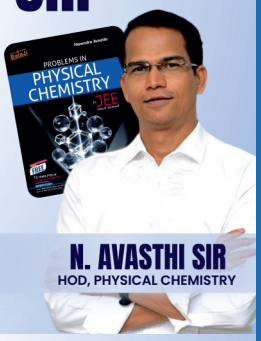

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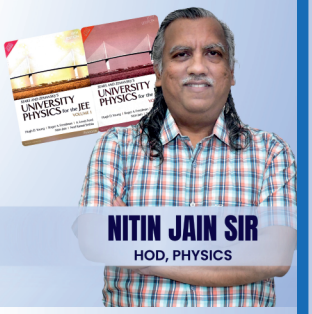


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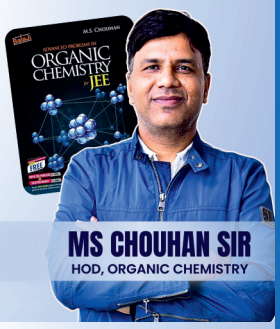


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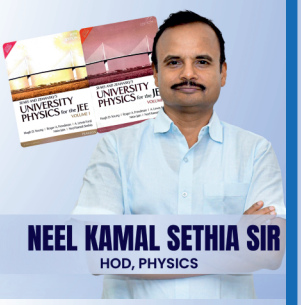


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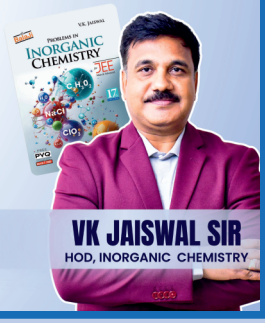


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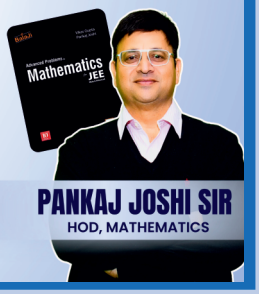


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MATHEMATICS

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MEGA-XII

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KEY CONCEPTS (LIMIT)

THINGS TO REMEMBER :

1. Limit of a function f(x) is said to exist as, $x \to a$ when

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = \text{finite quantity.}$$

2. FUNDAMENTAL THEOREMS ON LIMITS:

Let $\lim_{x \to a} f(x) = l$ & $\lim_{x \to a} g(x) = m$. If l & m exists then :

(i) $\lim_{x \to a} f(x) \pm g(x) = l \pm m$ **(ii)** $\lim_{x \to a} f(x) \cdot g(x) = l.m$

(iii) $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\ell}{m}$, provided $m \neq 0$

(iv) $\lim_{x \to a} k f(x) = k \lim_{x \to a} f(x)$; where k is a constant.

(v) $\lim_{x \to a} f[g(x)] = f\left(\lim_{x \to a} g(x)\right)$; provided f is continuous at $x = \lim_{x \to a} g(x)$

3. STANDARD LIMITS :

(a) $\lim_{x \to 0} \frac{\sin x}{x} = 1 = \lim_{x \to 0} \frac{\tan x}{x} = \lim_{x \to 0} \frac{\tan^{-1} x}{x} = \lim_{x \to 0} \frac{\sin^{-1} x}{x}$

[Where x is measured in radians]

(b) $\lim_{x \to 0} (1 + x)^{1/x} = e = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x$ note however there $\lim_{\substack{h \to 0 \\ n \to \infty}} (1 - h^2)^n = 0$

and $\lim_{\substack{h \to 0 \\ n \to \infty}} (1 + h^2)^n \to \infty$

(c) If $\lim_{x \to a} f(x) = 1$ and $\lim_{x \to a} \phi(x) = \infty$, then ;

$$\lim_{x \to a} [f(x)]^{\phi(x)} = e^{\lim_{x \to a} \phi(x)[f(x) - 1]}$$

(d) If $\lim_{x \to a} f(x) = A > 0$ & $\lim_{x \to a} \phi(x) = B$ (a finite quantity) then ;

$$\lim_{x \to a} [f(x)]^{\phi(x)} = A^B$$

(e) $\lim_{x \to 0} \frac{a^x - 1}{x} = l$n a (a > 0). In particular $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$

(f) $\lim_{x \to a} \frac{x^n - a^n}{x - a} = n a^{n-1}$

4. SQUEEZE PLAY THEOREM :

If $f(x) \leq g(x) \leq h(x)\ \forall\ x$ **&** $\lim_{x \to a} f(x) = l = \lim_{x \to a} h(x)$ then $\lim_{x \to a} g(x) = l$.

5. INDETERMINANT FORMS :

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty,\ 0^\circ,\ \infty^\circ, \infty - \infty \text{ and } 1^\infty$$

Note :

(i) We cannot plot ∞ on the paper. Infinity (∞) is a symbol & not a number. It does not obey the laws of elementry algebra.

(ii) $\infty + \infty = \infty$ **(iii)** $\infty \times \infty = \infty$ **(iv)** $(a/\infty) = 0$ if a is finite

(v) $\frac{a}{0}$ is not defined , if $a \neq 0$.

(vi) a b = 0 , if & only if a = 0 or b = 0 and a & b are finite.

6. The following strategies should be born in mind for evaluating the limits:

(a) Factorisation

(b) Rationalisation or double rationalisation

(c) Use of trigonometric transformation ;
appropriate substitution and using standard limits

(d) Expansion of function like Binomial expansion, exponential & logarithmic expansion, expansion of sinx , cosx , tanx should be remembered by heart & are given below :

(i) $a^x = 1 + \frac{x\,ln\,a}{1!} + \frac{x^2\,ln^2 a}{2!} + \frac{x^3 ln^3 a}{3!} + ...a > 0$ **(ii)** $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$

(iii) $ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...for -1 < x \leq 1$ **(iv)** $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ...$

(v) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ...$ **(vi)** $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + ...$

(vii) $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ...$

(viii) $\sin^{-1}x = x + \frac{1^2}{3!}x^3 + \frac{1^2.3^2}{5!}x^5 + \frac{1^2.3^2.5^2}{7!}x^7 + ...$

(ix) $\sec^{-1}x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + ...$

PROFICIENCY TEST-01

1. $\lim\limits_{x\to 1} \dfrac{x^2-\sqrt{x}}{\sqrt{x}-1}$ **2.** $\lim\limits_{x\to 1} \dfrac{x^2-x\ln x+\ln x-1}{x-1}$ **3.** $\lim\limits_{x\to 1} \dfrac{\sqrt[13]{x}-\sqrt[7]{x}}{\sqrt[5]{x}-\sqrt[3]{x}}$

4. $\lim\limits_{x\to 0}$ [*l*n (1 + sin²x). cot(*l*n² (1 + x))]

5. (a) $\lim\limits_{x\to 0} \tan^{-1}\dfrac{a}{x^2}$, where $a\in R$; (b) Plot the graph of the function f(x) = $\lim\limits_{t\to 0}\left(\dfrac{2x}{\pi}\tan^{-1}\dfrac{x}{t^2}\right)$

6. $\lim\limits_{x\to 1} \dfrac{\left[\sum\limits_{K=1}^{100} x^k\right]-100}{x-1}$

7. $\lim\limits_{x\to\infty} \dfrac{2\sqrt{x}+3x^{1/3}+5x^{1/5}}{\sqrt{3x-2}+(2x-3)^{1/3}}$

8. $\lim\limits_{x\to 0} \dfrac{\sec 4x-\sec 2x}{\sec 3x-\sec x}$

9. $\lim\limits_{x\to 0} \dfrac{8}{x^8}\left[1-\cos\dfrac{x^2}{2}-\cos\dfrac{x^2}{4}+\cos\dfrac{x^2}{2}\cos\dfrac{x^2}{4}\right]$

10. $\lim\limits_{\theta\to\frac{\pi}{4}} \dfrac{\sqrt{2}-\cos\theta-\sin\theta}{(4\theta-\pi)^2}$

PROFICIENCY TEST-02

1. If *I* = $\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}\left((r+1)\sin\dfrac{\pi}{r+1}-r\sin\dfrac{\pi}{r}\right)$ then find { *I* }. (where { } denotes the fractional part function)

2. $\lim\limits_{x\to-\infty} \dfrac{(3x^4+2x^2)\sin\frac{1}{x}+|x|^3+5}{|x|^3+|x|^2+|x|+1}$

3. Let $S_n = a_1 + a_2 + a_3 + ... + a_n$ where $\lim\limits_{n\to\infty} a_n = 2$, then find the value of $\lim\limits_{n\to\infty} \dfrac{S_{n+1}-S_n}{\sum\limits_{k=1}^{n} k}$

4. Calculate $\lim\limits_{x\to 0}\left(1+xe^{-\frac{1}{x^2}}\sin\dfrac{1}{x^4}\right)$

5. $\lim\limits_{x\to\infty}\left(\dfrac{x+c}{x-c}\right)^x = 4$ then find c

6. $\lim_{x\to\infty}\left[\frac{2x^2+3}{2x^2+5}\right]^{8x^2+3}$

7. $\lim_{n\to\infty}\left(\cos(l\,n(n-1)-l\,n(n+1))\right)^{(n+1)^2}$

8. Evaluate, $\lim_{x\to1}\frac{1-x+\ln x}{1+\cos\pi x}$

9. $\lim_{y\to0}\left[\underset{x\to\infty}{\text{Limit}}\frac{\exp\left(x\ln(1+\frac{ay}{x})\right)-\exp\left(x1n(1+\frac{by}{x})\right)}{y}\right]$

10. $\lim_{x\to1}\frac{ax^2+bx+c}{(x-1)^2}=2$, then find the value of $\lim_{x\to1}\frac{(x-a)(x-b)(x-c)}{(x+1)}$

EXERCISE–I

1. $\lim\limits_{x\to 1}\left(\frac{p}{1-x^p}-\frac{q}{1-x^q}\right)$ p, q ∈ N

2. If L = $\lim\limits_{x\to 1}\frac{(1-x)(1-x^2)(1-x^3).....(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3).....(1-x^n)]^2}$ then show that L can be equal to

(a) $\prod\limits_{r=1}^{n}\frac{n+r}{r}$ (b) $\frac{1}{n!}\prod\limits_{r=1}^{n}(4r-2)$

(c) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$.

(d) The coefficient of x^n in the expansion of $(1+x)^{2n}$.

3. $\lim\limits_{x\to\infty}$ (x – *l* n cosh x) where cosh t = $\frac{e^t+e^{-t}}{2}$.

4. $\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\cos^{-1}\left(2x\sqrt{1-x^2}\right)}{x-\frac{1}{\sqrt{2}}}$

5. $\lim\limits_{x\to\frac{\pi}{2}}\frac{2^{-\cos x}-1}{x(x-\frac{\pi}{2})}$

6. If $\lim\limits_{x\to 0}\frac{a\sin x-\sin 2x}{\tan^3 x}$ is finite then find the value of 'a' & the limit.

7. If $f_1(x)=\frac{x}{2}+10\ \forall\ x\in R$ and $f_n(x)=f_1(f_{n-1}(x))$, $\forall\ n\geq 2$, $n\in N$, then evaluate $\lim\limits_{n\to\infty} f_n(x)$

8. Let $f(x)=\frac{x}{\sin x},\ x>0$ $\quad=2-x,\ x\leq 0$ and $g(x)=x+3,\ x<1$ $\quad=x^2-2x-2,\ 1\leq x<2$ $\quad=x-5,\ x\geq 2$

find LHL and RHL of $g(f(x))$ at x = 0 and hence find $\lim\limits_{x\to 0} g(f(x))$.

9. Let $P_n=a^{P_{n-1}}-1$, $\forall$ n = 2, 3,.....and Let $P_1=a^x-1$ where a ∈ R^+ then evaluate $\lim\limits_{x\to 0}\frac{P_n}{x}$.

10. Determine $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sin\left(\frac{\pi/2}{k}\right)-\cos\left(\frac{\pi/2}{k}\right)-\sin\left(\frac{\pi/2}{k+2}\right)+\cos\left(\frac{\pi/2}{k+2}\right)\right)$.

11. $\lim\limits_{x\to\infty}\left(\frac{a_1^{\frac{1}{x}}+a_2^{\frac{1}{x}}+a_3^{\frac{1}{x}}+.....+a_n^{\frac{1}{x}}}{n}\right)^{nx}$ where $a_1, a_2, a_3, ...a_n > 0$

12. Let $L = \prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M = \prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ and $N = \prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$, then find the value of $L^{-1} + M^{-1} + N^{-1}$.

13. If $f(n, \theta)=\prod_{r=1}^{n}\left(1-\tan^2\frac{\theta}{2^r}\right)$, then compute $\lim_{n\to\infty} f(n,\theta)$

14. Using Sandwich theorem, evaluate

(a) $\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\ldots\ldots\ldots+\frac{1}{\sqrt{n^2+2n}}\right)$

(b) $\lim_{n\to\infty}\frac{1}{1+n^2}+\frac{2}{2+n^2}+\ldots\ldots\ldots+\frac{n}{n+n^2}$

(c) $\lim_{n\to\infty}\left(\sqrt{2}-2^{1/3}\right)\left(\sqrt{2}-2^{1/5}\right)\ldots\ldots\ldots\left(\sqrt{2}-2^{1/(2n+1)}\right)$ (d) $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{{}^nC_k}$

15. If $L = \lim_{x\to0}\left(\frac{1}{\ln(1+x)}-\frac{1}{\ln(x+\sqrt{1+x^2})}\right)$ then find the value of $\frac{L+153}{L}$.

EXERCISE–II

1. $\lim_{x\to1}\frac{(l\text{n}(1+x)-l\text{n}\,2)(3.4^{x-1}-3x)}{[(7+x)^{\frac{1}{3}}-(1+3x)^{\frac{1}{2}}].\sin(x-1)}$

2. (i) $\lim_{x\to2}\frac{(\cos\alpha)^x+(\sin\alpha)^x-1}{x-2}$ (ii) $\lim_{x\to4}\frac{(\cos\alpha)^x-(\sin\alpha)^x-\cos2\alpha}{x-4}$

3. If the $\lim_{x\to0}\frac{1}{x^3}\left(\frac{1}{\sqrt{1+x}}-\frac{1+ax}{1+bx}\right)$ exists and has the value equal to l, then find the value of $\frac{1}{a}-\frac{2}{l}+\frac{3}{b}$.

4. Find a & b if : (i) $\lim_{x\to\infty}\left[\frac{x^2+1}{x+1}-ax-b\right]=0$ (ii) $\lim_{x\to-\infty}\left[\sqrt{x^2-x+1}-ax-b\right]=0$

5. $\lim_{n\to\infty}\left(\frac{\sqrt{n^2+n}-1}{n}\right)^{2\sqrt{n^2+n}-1}$

6. $\lim_{x\to\infty}\left[\cos\left(2\pi\left(\frac{x}{1+x}\right)^a\right)\right]^{x^2}$ $a \in R$

7. Let f(x) = $\frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\,.\,(1-\{x\})}$ then find $\lim_{x\to 0^+}$ f(x) and $\lim_{x\to 0^-}$ f(x), where {x} denotes the fractional part function.

8. Find the values of a, b & c so that $\lim_{x\to 0}\frac{ae^x - b\cos x + ce^{-x}}{x.\sin x} = 2$

9. If a sequence of numbers $\{x_n\}$, determine by the equality $x_n = \frac{x_{n-1}+x_{n-2}}{2}$ and the values x_0 & x_1, Prove that $\lim_{n\to\infty} x_n = \frac{x_0+2x_1}{3}$.

10. If $\lim_{x\to 1}\left(1+ax+bx^2\right)^{\frac{a}{x-1}} = e^{-3}$, find the values of ordered pair (a, b).

11. $\lim_{n\to\infty}\frac{[1.x]+[2.x]+[3.x]+.....+[n.x]}{n^2}$, Where [.] denotes the greatest integer function.

12. Let *f* (x) = $\lim_{n\to\infty}\sum_{n=1}^{n} 3^{n-1}\sin^3\frac{x}{3^n}$ and g (x) = x – 4 *f* (x). Evaluate $\lim_{x\to 0}(1+g(x))^{\cot x}$.

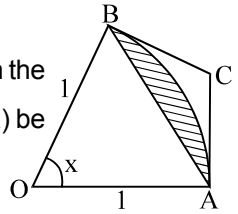
13. L = $\lim_{x\to 0}\frac{\sqrt{\frac{\cos 2x+(1+3x)^{1/3}}{2}} - \sqrt[3]{\frac{4\cos^3 x - \ln(1+x)^4}{4}}}{x}$

If L = a/b where 'a' and 'b' are relatively primes find (a + b).

14. $\lim_{x\to\infty}\left(\frac{\cosh(\pi/x)}{\cos(\pi/x)}\right)^{x^2}$ where cosh t = $\frac{e^t+e^{-t}}{2}$

15. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

16. A circular arc of radius 1 subtends an angle of x radians, $0 < x < \frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B. Let T(x) be the area of triangle ABC & let S(x) be the area of the shaded region. Compute:



(a) T(x) (b) S(x) & (c) the limit of $\frac{T(x)}{S(x)}$ as $x \to 0$.

17. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

EXERCISE–III

1. $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{1-\tan x}{1-\sqrt{2}\sin x}$

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

2. Find the sum of an infinite geometric series whose first term is the limit of the function

$f(x) = \dfrac{\tan x - \sin x}{\sin^3 x}$ as $x \to 0$ and whose common ratio is the limit of the function

$g(x) = \dfrac{1-\sqrt{x}}{(\cos^{-1} x)^2}$ as $x \to 1$.

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) $\frac{3}{5}$ (D) $\frac{5}{3}$

3. $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{\sqrt{1-\sqrt{\sin 2x}}}{\pi-4x}$

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{4}{\pi}$ (D) does not exist

4. $\lim\limits_{x\to 0} \dfrac{27^x - 9^x - 3^x + 1}{\sqrt{2}-\sqrt{1+\cos x}}$

(A) $4\sqrt{2}(\ln 3)^2$ (B) $8\sqrt{2}(\ln 3)^2$ (C) $4\sqrt{2}(\ln 3)$ (D) $8\sqrt{2}(\ln 3)$

5. $\lim\limits_{x\to 0} \left[\dfrac{(1+x)^{1/x}}{e}\right]^{1/x}$

(A) e (B) $\frac{1}{e}$ (C) $e^{-1/2}$ (D) $\sqrt{e}$

6. $\lim\limits_{x\to 0} \left[\dfrac{x^2}{\sin x \tan x}\right]$, where [.] denotes greatest integer function.

(A) 0 (B) –1 (C) 1 (D) does not exist

7. $\lim\limits_{x\to 1} \left(\tan\dfrac{\pi x}{4}\right)^{\tan\frac{\pi x}{2}}$

(A) e (B) $\frac{1}{e}$ (C) $e^{-1/2}$ (D) $\sqrt{e}$

8. Let $x_0 = 2\cos\dfrac{\pi}{6}$ and $x_n = \sqrt{2+x_{n-1}}$, n = 1, 2, 3, ..., find $\lim\limits_{n\to\infty} 2^{(n+1)}\cdot\sqrt{2-x_n}$

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

9. $\lim\limits_{x\to 0} \dfrac{2(\tan x-\sin x)-x^3}{x^5}$

(A) $-\dfrac{1}{2}$ (B) $\dfrac{2}{3}$ (C) $\dfrac{1}{4}$ (D) $\dfrac{5}{6}$

10. Calculate $\lim\limits_{x\to 0} \left(1+e^{-\frac{1}{x^2}} \arctan\dfrac{1}{x^2}+xe^{-\frac{1}{x^2}} \sin\dfrac{1}{x^4}\right)^{e^{\frac{1}{x^2}}}$

(A) e (B) e^2 (C) $e^{\pi/2}$ (D) $e^{-\pi/2}$

11. If $f(x) = \begin{cases} \sin x\ , & x \neq n\pi\ ,\ n \in I \\ 2\ , & \text{otherwise} \end{cases}$ and $g(x) = \begin{cases} x^2+1\ , & x \neq 0 \\ 4\ , & x = 0 \\ 6\ , & x = 2 \end{cases}$, then $\lim\limits_{x\to 0}$ g{f(x)} is :

(A) 1 (B) 4 (C) 5 (D) 6

12. $\lim\limits_{n\to\infty} \left(1+\dfrac{1}{5}\right)\left(1+\dfrac{1}{5^2}\right)\left(1+\dfrac{1}{5^4}\right)\ldots\ldots\left(1+\dfrac{1}{5^{2^n}}\right)$ is equal to :

(A) 0 (B) 5/4 (C) 4/5 (D) 1/5

13. The value of the limit $\lim\limits_{n\to\infty} n^2\left(a^{1/n}-a^{1/(n+1)}\right)$ $(a > 0)$ is :

(A) ln a (B) e^a (C) e^{-a} (D) 1

14. $\lim\limits_{n\to\infty} \dfrac{1^6+2^6+3^6+\ldots\ldots+n^6}{(1^2+2^2+3^2+\ldots\ldots+n^2)(1^3+2^3+3^3+\ldots\ldots+n^3)}$ =

(A) 2 (B) $\dfrac{21}{8}$ (C) 1 (D) $\dfrac{12}{7}$

15. If a, b, c are non zero constant numbers then $\lim\limits_{r\to\infty} \dfrac{\cos\frac{a}{r}-\cos\frac{b}{r}\cos\frac{c}{r}}{\sin\frac{b}{r}\sin\frac{c}{r}}$ equals :

(A) $\dfrac{a^2+b^2-c^2}{2bc}$ (B) $\dfrac{c^2+a^2-b^2}{2bc}$ (C) $\dfrac{b^2+c^2-a^2}{2bc}$ (D) Independent of a, b, c

EXERCISE–IV(Mains)

1. If $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b are **[AIEEE 2004]**

(A) $a \in R, b \in R$ (B) $a = 1, b \in R$ (C) $a \in R, b = 2$ (D) $a = 1, b = 2$

2. Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim\limits_{x\to\alpha}\frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$ is equal to

[AIEEE 2005]

(A) $\frac{a^2}{2}(\alpha-\beta)^2$ (B) 0 (C) $\frac{a}{2}(\alpha-\beta)^2$ (D) $\frac{1}{2}(\alpha-\beta)^2$

3. Let $f : R \to R$ be a positive, increasing function with $\lim\limits_{x\to\infty}\frac{f(3x)}{f(x)}=1$. Then $\lim\limits_{x\to\infty}\frac{f(2x)}{f(x)}$ is equal to

[AIEEE 2010]

(A) 3 (B) 1 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

4. $\lim\limits_{x\to2}\left(\frac{\sqrt{1-\cos\{2(x-2)\}}}{x-2}\right)$ **[AIEEE 2011]**

(A) Equals $\frac{1}{\sqrt{2}}$ (B) Does not exist (C) Equals $\sqrt{2}$ (D) Equals $-\sqrt{2}$

5. $\lim\limits_{x\to0}\frac{\sin(\pi\cos^2 x)}{x^2}$ is equal to **[IIT Mains - 2014]**

(A) π (B) $\frac{\pi}{2}$ (C) 1 (D) $-\pi$

6. Let f(x) be a polynomial of degree four having extreme values at x = 1 and x=2. If $\lim\limits_{x\to0}\left[1+\frac{f(x)}{x^2}\right]=3$, then f(2) is equal to: **[IIT Mains - 2015]**

(A) 4 (B) –8 (C) –4 (D) 0

7. $\lim\limits_{x\to0}\frac{(1-\cos2x)(3+\cos x)}{x\tan4x}$ is equal to : **[IIT Mains - 2015]**

(A) $\frac{1}{2}$ (B) 4 (C) 3 (D) 2

8. Let $p = \lim\limits_{x\to0^+}\left(1+\tan^2\sqrt{x}\right)^{\frac{1}{2x}}$ then log p is equal to : **[IIT Mains-2016]**

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

9. $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{\cot x - \cos x}{(\pi - 2x)^3}$ equals : **[IIT Mains - 2017]**

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{24}$ (D) $\frac{1}{16}$

10. For each t $\in$ R, let [t] be the greatest integer less than or equal to t. Then $\lim\limits_{x\to 0} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+.....+\left[\frac{15}{x}\right]\right)$

(A) does not exist (in R) (B) is equal to 0 **[IIT Mains - 2018]**

(C) is equal to 15 (D) is equal to 120

EXERCISE–V(Advanced)

1. $\underset{x\to 0}{\text{Lim}} \dfrac{x\tan 2x - 2x\tan x}{(1-\cos 2x)^2}$ is : [JEE '99, 2 (out of 200)]

(A) 2 (B) -2 (C) 1/2 (D) $-1/2$

2. For x $\in$ R, $\underset{x\to\infty}{\text{Lim}} \left(\dfrac{x-3}{x+2}\right)^x$ = [JEE 2000, Screening]

(A) e (B) e^{-1} (C) e^{-5} (D) e^5

3. Evaluate $\underset{x\to 0}{\text{Lim}} \dfrac{a^{\tan x} - a^{\sin x}}{\tan x - \sin x}$, a > 0. [REE 2001, 3 out of 100]

4. The integer n for which $\underset{x\to 0}{\text{Lim}} \dfrac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number is

(A) 1 (B) 2 (C) 3 (D) 4

[JEE 2002 (screening), 3]

5. If $\underset{x\to 0}{\text{Lim}} \dfrac{\sin(nx)[(a-n)nx - \tan x]}{x^2} = 0$ (n > 0) then the value of 'a' is equal to [JEE 2003 (screening)]

(A) $\frac{1}{n}$ (B) $n^2 + 1$ (C) $\frac{n^2+1}{n}$ (D) None

6. Find the value of $\underset{n\to\infty}{\text{Lim}} \left[\frac{2}{\pi}(n+1)\cos^{-1}\left(\frac{1}{n}\right) - n\right]$. [JEE ' 2004, 2 out of 60]

7. Let L = $\lim\limits_{x\to 0} \dfrac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, a > 0. If L is finite, then [JEE ' 2009, 3]

(A) a = 2 (B) a = 1 (C) L = $\frac{1}{64}$ (D) L = $\frac{1}{32}$

8. If $\lim_{x\to 0}\left[1+x\ln(1+b^2)\right]^{1/x} = 2b\sin^2\theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is [JEE ' 2011]

(A) $\pm\frac{\pi}{4}$ (B) $\pm\frac{\pi}{3}$ (C) $\pm\frac{\pi}{6}$ (D) $\pm\frac{\pi}{2}$

9. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation [JEE ' 2012]

$\left(\sqrt[3]{1+a}-1\right)x^2+\left(\sqrt{1+a}-1\right)x+\left(\sqrt[6]{1+a}-1\right)=0$ where $a > -1$.

Then $\lim_{a\to 0^+}\alpha(a)$ and $\lim_{a\to 0^+}\beta(a)$ are

(A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1 (C) $-\frac{7}{2}$ and 2 (D) $-\frac{9}{2}$ and 3

10. If $\lim_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$, then [JEE ' 2012]

(A) $a = 1, b = 4$ (B) $a = 1, b = -4$ (C) $a = 2, b = -3$ (D) $a = 2, b = 3$

11. The largest value of the non-negative integer a for which $\lim_{x\to 1}\left\{\frac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}}=\frac{1}{4}$ is

[IIT Advanced-2014]

12. Let m and n be two positive integers greater than 1. If $\lim_{\alpha\to 0}\left(\frac{e^{\cos(a^n)}-e}{\alpha^m}\right)=-\left(\frac{e}{2}\right)$ then the value of $\frac{m}{n}$ is

[IIT Advanced - 2015]

13. Let $\alpha, \beta \in \mathbf{R}$ be such that $\lim_{x\to 0}\frac{x^2\sin(\beta x)}{\alpha x-\sin x}=1$. Then $6(\alpha+\beta)$ equals : **[IIT Advanced - 2016]**

14. Let $f(x)=\frac{1-x(1+|1-x|)}{|1-x|}\cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then **[IIT Advanced - 2017]**

(A) $\lim_{x\to 1^+} f(x)$ does not exist (B) $\lim_{x\to 1^-} f(x)$ does not exist

(C) $\lim_{x\to 1^-} f(x) = 0$ (D) $\lim_{x\to 1^+} f(x) = 0$

15. For any positive integer n, define $f_n : (0, \infty) \to \mathbb{R}$ as **[IIT Advanced - 2018]**

$$f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right) \text{ for all } x \in (0, \infty).$$

(Here, the inverse trigonometric function $\tan^{-1} x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

Then, which of the following statement(s) is (are) TRUE?

(A) $\sum_{j=1}^{5}\tan^2(f_j(0)) = 55$ (B) $\sum_{j=1}^{10}(1+f_j'(0))\sec^2(f_j(0)) = 10$

(C) For any fixed positive integer n, $\lim_{x\to\infty}\tan(f_n(x))=\frac{1}{n}$

(D) For any fixed positive integer n, $\lim_{x\to\infty}\sec^2(f_n(x)) = 1$

16. For any positive integer n, let $S_n : (0, \infty) \to R$ be defined by

$$S_n(x) = \sum_{k=1}^{n} \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$$

where for any $x \in R$, $\cot^{-1}(x) \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) **TRUE**? **[IIT Advanced - 2021]**

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x > 0$ (B) $\lim_{n\to\infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$ (D) $\tan(S_n(x)) \le \frac{1}{2}$, for all $n \ge 1$ and $x > 0$

17. If $\beta = \lim_{x\to 0} \frac{e^{x^3} - (1-x^3)^{\frac{1}{3}} + ((1-x^2)^{\frac{1}{2}} - 1)\sin x}{x\sin^2 x}$

then the value of 6β is _____. **[IIT Advanced - 2022]**

18. Let $k \in \mathbb{R}$. If $\lim_{x\to 0^+} \left(\sin(\sin kx) + \cos x + x\right)^{\frac{2}{x}} = e^6$, then the value of k is. **[IIT Advanced - 2024]**

(A) 1 (B) 2 (C) 3 (D) 4

19. Let S be the set of all $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ such that $\lim_{x\to\infty} \frac{\sin(x^2)(\log_e x)^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\log_e(1+x))^\beta} = 0$.

Then which of the following is (are) correct? **[IIT Advanced - 2024]**

(A) $(-1, 3) \in S$ (B) $(-1, 1) \in S$ (C) $(1, -1) \in S$ (D) $(1, -2) \in S$

20. Let x_0 be the real number such that $e^{x_0} + x_0 = 0$. For a given real number α, define

$$g(x) = \frac{3xe^x + 3x - \alpha e^x - \alpha x}{3(e^x + 1)} \text{ for all real numbers } x.$$

Then which one of the following statements is TRUE? **[IIT Advanced - 2025]**

(A) For $\alpha = 2$, $\lim_{x\to x_0} \left|\frac{g(x) + e^{x_0}}{x - x_0}\right| = 0$ (B) For $\alpha = 2$, $\lim_{x\to x_0} \left|\frac{g(x) + e^{x_0}}{x - x_0}\right| = 1$

(C) For $\alpha = 3$, $\lim_{x\to x_0} \left|\frac{g(x) + e^{x_0}}{x - x_0}\right| = 0$ (D) For $\alpha = 3$, $\lim_{x\to x_0} \left|\frac{g(x) + e^{x_0}}{x - x_0}\right| = \frac{2}{3}$

ANSWER KEY

PROFICIENCY TEST-01

1. 3 **2.** 2 **3.** $\frac{45}{91}$ **4.** 1

5. (a) $\pi/2$ if a > 0 ; 0 if a = 0 and $-\pi/2$ if a < 0 (b) f(x) = | x | **6.** 5050

7. $\frac{2}{\sqrt{3}}$ **8.** $\frac{3}{2}$ **9.** $\frac{1}{32}$ **10.** $\frac{1}{16\sqrt{2}}$

PROFICIENCY TEST-02

1. $\pi - 3$ **2.** -2 **3.** 0 **4.** 1 **5.** c = ln2 **6.** e^{-8} **7.** e^{-2}

8. $-\frac{1}{\pi^2}$ **9.** a – b **10.** 5/2

EXERCISE–I

1. $\frac{p-q}{2}$ **3.** ln 2 **4.** does not exist **5.** $\frac{2\ln 2}{\pi}$ **6.** a = 2 ; limit = 1 **7.** 20

8. – 3, –3, – 3 **9.** $(l\text{n } a)^n$ **10.** 3 **11.** $(a_1.a_2.a_3....a_n)$ **12.** 8

13. $\frac{\theta}{\tan\theta}$ **14.** (a) 2 (b) 1/2 (c) 0 (d) 2 **15.** 307

EXERCISE–II

1. $-\frac{9}{4}\,l\text{n}\,\frac{4}{e}$ **2.** (i) $\cos^2\alpha$ lncosα + $\sin^2\alpha$ lnsinα, (ii) $\cos^4\alpha$ lncosα – $\sin^4\alpha$ lnsinα

3. 72 **4.** **(i)** a =1, b = –1 **(ii)** a = –1 , b =1/2 **5.** e^{-1} **6.** $e^{-2\pi^2 a^2}$ **7.** $\frac{\pi}{2}, \frac{\pi}{2\sqrt{2}}$

8. a = c = 1, b = 2 **10.** $\left(\sqrt{3},-\sqrt{3}\right), \left(-\sqrt{3},\sqrt{3}\right)$ **11.** $\frac{x}{2}$

12. g(x) = sin x and limit = e **13.** 19 **14.** e^{π^2}

16. T(x) = $\frac{1}{2}\tan^2\frac{x}{2}$. sin x or $\tan\frac{x}{2} - \frac{\sin x}{2}$, S(x) = $\frac{1}{2}x - \frac{1}{2}\sin x$, limit = $\frac{3}{2}$

EXERCISE–III

1. D **2.** A **3.** D **4.** B **5.** C **6.** A **7.** B
8. D **9.** C **10.** C **11.** A **12.** B **13.** A **14.** D
15. C

EXERCISE–IV(Mains)

1. B **2.** A **3.** B **4.** B **5.** A **6.** D **7.** D
8. C **9.** D **10.** D

EXERCISE–V(Advanced)

1. C **2.** C **3.** lna **4.** C **5.** C **6.** $1-\frac{2}{\pi}$ **7.** A,C
8. D **9.** B **10.** B **11.** 0 **12.** 2 **13.** 7 **14.** A, C
15. D **16.** A,B **17.** 5 **18.** B **19.** B, C **20.** C

CONTINUITY
KEY CONCEPTS

THINGS TO REMEMBER :

1. A function f(x) is said to be continuous at x = a if $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$ i.e.

L.H.L. = R.H.L. = value of the function at x = a i.e. $\lim_{x\to a} f(x) = f(a)$

If f(x) is not continuous at x = a, we say that f(x) is discontinuous at x = a.
f(x) will be discontinuous at x = a in any of the following cases.

(i) $\lim_{x\to a} f(x)$ does not exist

(ii) $\lim_{x\to a} f(x) \neq f(a)$

(iii) f(a) is not defined.

2. Types of Discontinuities :

Type - 1: (Removable type of discontinuities)

In case $\underset{x\to c}{\text{Limit}}\ f(x)$ exists but is not equal to f(c) then the function is said to have a removable discontinuity or discontinuity of the first kind. In this case we can redefine the function such that $\underset{x\to c}{\text{Limit}}\ f(x) = f(c)$ & make it continuous at x= c. Removable type of discontinuity can be further classified as :

(a) MISSING POINT DISCONTINUITY : Where $\underset{x\to a}{\text{Limit}}\ f(x)$ exists finitely but f(a) is not defined.

e.g. $f(x) = \frac{(1-x)(9-x^2)}{(1-x)}$ has a missing point discontinuity at x = 1 , and $f(x) = \frac{\sin x}{x}$ has a missing point discontinuity at x = 0

(b) ISOLATED POINT DISCONTINUITY : Where $\underset{x\to a}{\text{Limit}}\ f(x)$ exists & f(a) also exists but ; $\underset{x\to a}{\text{Limit}} \neq f(a)$.

e.g. $f(x) = \frac{x^2-16}{x-4}$, $x \neq 4$ & f (4) = 9 has an isolated point discontinuity at x = 4.

Similarly f(x) = [x] + [–x] = $\begin{cases} 0 & \text{if } x \in I \\ -1 & \text{if } x \notin I \end{cases}$ has an isolated point discontinuity at all $x \in I$.

Type-2: (Non - Removable type of discontinuities)

In case $\underset{x\to c}{\text{Limit}}\ f(x)$ does not exist then it is not possible to make the function continuous by redefining it. Such discontinuities are known as non - removable discontinuity or discontinuity of the 2nd kind. Non-removable type of discontinuity can be further classified as :

(a) Finite discontinuity e.g. f(x) = x – [x] at all integral x ; $f(x) = \tan^{-1}\frac{1}{x}$ at x = 0 and $f(x) = \frac{1}{1+2^{\frac{1}{x}}}$ at x = 0 (note that $f(0^+) = 0$; $f(0^-) = 1$)

(b) Infinite discontinuity e.g. $f(x) = \frac{1}{x-4}$ or $g(x) = \frac{1}{(x-4)^2}$ at x = 4 ; $f(x) = 2^{\tan x}$ at $x = \frac{\pi}{2}$ and $f(x) = \frac{\cos x}{x}$ at x = 0.

(c) Oscillatory discontinuity e.g. $f(x) = \sin\frac{1}{x}$ at x = 0.

In all these cases the value of f(a) of the function at x= a (point of discontinuity) may or may not exist but $\underset{x\to a}{\text{Limit}}\ f(x)$ does not exist.

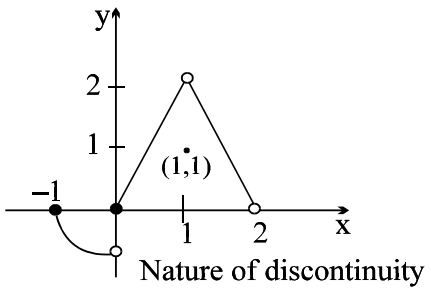
Note: From the adjacent graph note that

– f is continuous at x = – 1

– f has isolated discontinuity at x = 1

– f has missing point discontinuity at x = 2

– f has non removable (finite type) discontinuity at the origin.

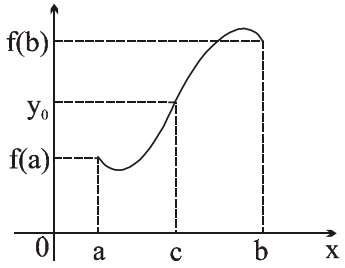


Nature of discontinuity

3. In case of dis-continuity of the second kind the non-negative difference between the value of the RHL at x = c & LHL at x = c is called **THE JUMP OF DISCONTINUITY.** A function having a finite number of jumps in a given interval I is called a **PIECE WISE CONTINUOUS** or **SECTIONALLY CONTINUOUS** function in this interval.

4. All Polynomials, Trigonometrical functions, exponential & Logarithmic functions are continuous in their domains.

5. If f & g are two functions that are continuous at x= c then the functions defined by :
$F_1(x) = f(x) \pm g(x)$; $F_2(x) = K f(x)$, K any real number; $F_3(x) = f(x).g(x)$ are also continuous at x= c. Further, if g (c) is not zero, then $F_4(x) = \frac{f(x)}{g(x)}$ is also continuous at x= c.



6. **The intermediate value theorem:**
Suppose f(x) is continuous on an interval I , and a and b are any two points of I. Then if y_0 is a number between f(a) and f(b) , their exists a number c between a and b such that $f(c) = y_0$.

NOTE VERY CAREFULLY THAT :

(a) If f(x) is continuous & g(x) is discontinuous at x = a then the product function $\phi(x) = f(x) \cdot g(x)$ is not necessarily be discontinuous at x = a. e.g.

$$f(x) = x \;\&\; g(x) = \begin{cases} \sin\frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

(b) If f(x) and g(x) both are discontinuous at x = a then the product function $\phi(x) = f(x) \cdot g(x)$ is not necessarily be discontinuous at x = a. e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

(c) A Continuous function whose domain is closed must have a range also in closed interval.

(d) If f is continuous at x = c & g is continuous at x = f(c) then the composite g[f(x)] is continuous at x = c. eg. $f(x) = \frac{x \sin x}{x^2 + 2}$ & $g(x) = |x|$ are continuous at x = 0 , hence the composite (gof) (x) = $\left|\frac{x \sin x}{x^2 + 2}\right|$ will also be continuous at x = 0 .

7. **CONTINUITY IN AN INTERVAL :**

(a) A function f is said to be continuous in (a , b) if f is continuous at each & every point $\in$(a , b).

(b) A function f is said to be continuous in a closed interval $[a,b]$ if :

(i) f is continuous in the open interval (a , b) &

(ii) f is right continuous at **'a'** i.e. $\lim_{x \to a^+} f(x) = f(a)$ = a finite quantity.

(iii) f is left continuous at **'b'** i.e. $\lim_{x \to b^-} f(x) = f(b)$ = a finite quantity.

Note that a function f which is continuous in $[a,b]$ possesses the following properties :

(i) If f(a) & f(b) possess opposite signs, then there exists at least one solution of the equation f(x) = 0 in the open interval (a , b).

(ii) If K is any real number between f(a) & f(b), then there exists at least one solution of the equation f(x) = K in the open inetrval (a , b).

PROFICIENCY TEST

1. If the function $f(x) = \dfrac{3x^2 + ax + a + 3}{x^2 + x - 2}$ is continuous at x = – 2. Find $f(-2)$.

2. Find all possible values of *a* and *b* so that $f(x)$ is continuous for all $x \in R$ if

$$f(x) = \begin{cases} |ax+3| & \text{if } x \le -1 \\ |3x+a| & \text{if } -1 < x \le 0 \\ \dfrac{b\sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \ge \pi \end{cases}$$

3. $f(x) = \begin{cases} ax^2 + bx + c, & |x| > 1 \\ x + 3, & |x| \le 1 \end{cases}$. Find the values of a, b, c so that f(x) is continuous for all values of x.

4. Draw the graph of the function $f(x) = x - |x - x^2|$, $-1 \le x \le 1$ & discuss the continuity or discontinuity of f in the interval $-1 \le x \le 1$.

5. Let $f(x) = \begin{cases} \frac{1-\sin \pi x}{1+\cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases}$. Determine the value of p, if possible, so that the function is continuous at x=1/2.

6. Given the function $g(x) = \sqrt{6-2x}$ and $h(x) = 2x^2 - 3x + a$. Then

(a) evaluate $h(g(2))$ (b) If $f(x) = \begin{cases} g(x), & x \le 1 \\ h(x), & x > 1 \end{cases}$, find 'a' so that f is continuous.

7. Let $f(x) = \begin{cases} 1+x, & 0 \le x \le 2 \\ 3-x, & 2 < x \le 3 \end{cases}$. Determine the form of $g(x) = f[f(x)]$ & hence find the point of discontinuity of g , if any.

8. Let [x] denote the greatest integer function & f(x) be defined in a neighbourhood of 2 by

$$f(x) = \begin{cases} \dfrac{\left(\exp\{(x+2)\ell n 4\}\right)^{\frac{[x+1]}{4}} - 16}{4^x - 16}, & x < 2 \\ A\dfrac{1-\cos(x-2)}{(x-2)\tan(x-2)}, & x > 2 \end{cases}.$$

Find the values of A & f(2) in order that f(x) may be continuous at x = 2.

9. Discuss the continuity of the function 'f' defined as follows: $f(x) = \begin{cases} \frac{1}{x-1} & \text{for} \quad 0 \le x \le 2 \\ \frac{3}{x+1} & \text{for} \quad 2 < x \le 4 \\ \frac{x+1}{x-5} & \text{for} \quad 4 < x \le 6 \end{cases}$ and draw the graph of the function for $x \in [0, 6]$. Also indicate the nature of discontinuities if any.

10. State whether True or False.

(i) $f(x) = \underset{n\to\infty}{\text{Limit}} \frac{1}{1+n\sin^2 \pi x}$ is continuous at x = 1.

(ii) The function defined by $f(x) = \frac{x}{|x|+2x^2}$ for $x \neq 0$ & f(0) = 1 is continuous at x = 0.

(iii) The function $f(x) = 2^{-2^{1/(1-x)}}$ if $x \neq 1$ & f(1) = 1 is not continuous at x = 1.

(iv) There exists a continuous function f: [0, 1] onto [0, 10], but there exists no continuous function g : [0, 1] onto (0, 10).

(v) If *f* (x) is continuous in [0 , 1] & f(x) = 1 for all rational numbers in [0 , 1] then f $\left(1/\sqrt{2}\right)$ equal to 1.

(vi) If *f* (x) = $\begin{cases} \frac{\cos \pi x + \sin(\pi x/2)}{(x-1)(3x^2-2x-1)} & \text{if } x \neq 1 \\ k & \text{if } k = 1 \end{cases}$ is continuous, then the value of *k* is $\frac{3\pi^2}{32}$.

EXERCISE-I

1. Let $f(x) = \begin{cases} \dfrac{\ln\cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x>0 \\ \dfrac{e^{\sin 4x}-1}{\ln(1+\tan 2x)} & \text{if } x<0 \end{cases}$

Is it possible to define f(0) to make the function continuous at x = 0. If yes what is the value of f(0), if not then indicate the nature of discontinuity.

2. Let $y_n(x) = x^2 + \dfrac{x^2}{1+x^2} + \dfrac{x^2}{(1+x^2)^2} + \ldots\ldots\ldots\ldots + \dfrac{x^2}{(1+x^2)^{n-1}}$ and $y(x) = \lim\limits_{n\to\infty} y_n(x)$

Discuss the continuity of $y_n(x)$ ($n \in N$) and $y(x)$ at x = 0

3. The function $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0<x<\frac{\pi}{2} \\ b+2 & \text{if } x=\frac{\pi}{2} \\ \left(1+|\cos x|\right)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2}<x<\pi \end{cases}$

Determine the values of 'a' & 'b' , if f is continuous at $x = \pi/2$.

4. $f(x) = \begin{cases} ax+b, & -1 \le x < 0 \\ \left[\dfrac{e^x-1}{x}\right], & 0 < x \le 1 \\ bx+a, & 1 < x \le 2 \end{cases}$. Where [.] denotes the greatest integer function.

Find values of 'a' and 'b' so that f(x) is continuous for all values of $x \in [-1, 2]$

5. Let $f(x) = \begin{cases} 1+x^3 & , x<0 \\ x^2-1 & , x \ge 0 \end{cases}$; $g(x) = \begin{cases} (x-1)^{1/3} & , x<0 \\ (x+1)^{1/2} & , x \ge 0 \end{cases}$. Discuss the continuity of g (f (x)).

6. Determine a & b so that f is continuous at $x = \dfrac{\pi}{2}$. $f(x) = \begin{cases} \frac{1-\sin^3 x}{3\cos^2 x} & \text{if } x<\frac{\pi}{2} \\ a & \text{if } x=\frac{\pi}{2} \\ \frac{b(1-\sin x)}{(\pi-2x)^2} & \text{if } x>\frac{\pi}{2} \end{cases}$

7. Determine the values of a, b & c for which the function $f(x) = \begin{cases} \frac{\sin(a+1)x+\sin x}{x} & \text{for } x<0 \\ c & \text{for } x=0 \\ \frac{\left(x+bx^2\right)^{1/2}-x^{1/2}}{bx^{3/2}} & \text{for } x>0 \end{cases}$

is continuous at x = 0.

8. If $f(x) = \frac{\sin 3x + A\sin 2x + B\sin x}{x^5}$ ($x \neq 0$) is cont. at x = 0. Find A & B. Also find f(0).

Select the correct alternative : (More than one are correct)

9. A function *f* is defined on an interval [a, b]. Which of the following statement(s) is/are INCORRECT?

(A) If *f* (a) and *f* (b), have opposite sign, then there must be a point $c \in (a, b)$ such that *f* (c) = 0.

(B) If *f* is continuous on [a, b], *f* (a) < 0 and *f* (b) > 0, then there must be a point $c \in (a, b)$ such that *f* (c) = 0

(C) If *f* is continuous on [a, b] and there is a point c in (a, b) such that *f* (c) = 0, then *f* (a) and *f* (b) have opposite sign.

(D) If *f* has no zeroes on [a, b], then *f* (a) and *f* (b) have the same sign.

10. Which of the following functions f has/have a removable discontinuity at the indicated point?

(A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$ at x = – 2

(B) $f(x) = \frac{x - 7}{|x - 7|}$ at x = 7

(C) $f(x) = \frac{x^3 + 64}{x + 4}$ at x = – 4

(D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$ at x = 9

11. Let 'f' be a continuous function on R. If $f(1/4^n) = \left(\sin e^n\right)e^{-n^2} + \frac{n^2}{n^2 + 1}$ then f(0) is :

(A) not unique

(B) 1

(C) data sufficient to find f(0)

(D) data insufficient to find f(0)

12. Indicate all correct alternatives if, $f(x) = \frac{x}{2} - 1$, then on the interval $[0, \pi]$

(A) tan (f (x)) & $\frac{1}{f(x)}$ are both continuous

(B) tan (f (x)) & $\frac{1}{f(x)}$ are both discontinuous

(C) tan (f (x)) & $f^{-1}(x)$ are both continuous

(D) tan (f (x)) is continuous but $\frac{1}{f(x)}$ is not

13. Let $f(x) = \begin{cases} \sin\left(\frac{5}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, then (fg)(x) is continuous at x = 0, if g(x) is equal to

(A) $(x^2 + 1), x \in R$ (B) $\ell n\,(1 + x), x \in (-1, \infty)$ (C) $\cos x, x \in R$ (D) $e^x, x \in R$

14. If $f(x) = \frac{1}{2 - x}$ and $g(x) = \frac{f(f(x))}{3f(f(x)) - 1}$, then the value(s) of x at which g(x) is discontinuous is/are

(A) 1 (B) 3/2 (C) 2 (D) 3

15. Let $f(x) = \begin{cases} x^2 + px + 1 \; ; \; x \in Q \\ px^2 + 2x + r \; ; \; x \notin Q \end{cases}$ and f(x) is continuous at x = 1 and x = 2, then

(A) p = 2 (B) r = 0 (C) p = 1/2 (D) r = 2

EXERCISE-II

1. If f(x) = x + {-x} + [x] , where [x] is the integral part & {x} is the fractional part of x. Discuss the continuity of f in [– 2, 2].

2. Find the locus of (a, b) for which the function f (x) = $\begin{cases} ax-b & \text{for} \quad x \le 1 \\ 3x & \text{for} \quad 1 < x < 2 \\ bx^2-a & \text{for} \quad x \ge 2 \end{cases}$ is continuous at x = 1 but discontinuous at x = 2.

3. Let g (x) = $\lim\limits_{n\to\infty} \dfrac{x^n f(x)+h(x)+1}{2x^n+3x+3}$, $x \neq 1$ and g (1) = $\lim\limits_{x\to 1} \dfrac{\sin^2(\pi\cdot 2^x)}{l n\left(\sec(\pi\cdot 2^x)\right)}$ be a continuous function at x = 1, find the value of 4 g (1) + 2 f (1) – h (1). Assume that f (x) and h (x) are continuous at x = 1.

4. If g : [a, b] onto [a, b] is continous show that there is some $c \in [a, b]$ such that g (c) = c.

5. The function f(x) = $\left(\dfrac{2+\cos x}{x^3 \sin x}-\dfrac{3}{x^4}\right)$ is not defined at x = 0. How should the function be defined at x = 0 to make it continuous at x = 0.

6. f (x) = $\dfrac{a^{\sin x}-a^{\tan x}}{\tan x-\sin x}$ for x > 0

= $\dfrac{l n(1+x+x^2)+l n(1-x+x^2)}{\sec x-\cos x}$ for x < 0, if f is continuous at x = 0, find 'a'.

Now if g (x) = $l n\left(2-\dfrac{x}{a}\right)\cdot \cot(x-a)$ for $x \neq a$, $a \neq 0$, a > 0. If g is continuous at x = a then show that $g(e^{-1}) = -e$.

7. (a) Let f(x + y) = f(x) + f(y) for all x , y & if the function f(x) is continuous at x = 0 , then show that f(x) is continuous at all x.

(b) If f(x· y) = f(x)· f(y) for all x , y and f(x) is continuous at x = 1. Prove that f(x) is continuous for all x except at x = 0. Given $f(1) \neq 0$.

8. Given f (x) = $\sum\limits_{r=1}^{n} \tan\left(\dfrac{x}{2^r}\right) \sec\left(\dfrac{x}{2^{r-1}}\right)$; r , n ∈ N

g (x) = $\underset{n\to\infty}{\text{Limit}} \dfrac{\ell n\left(f(x)+\tan\frac{x}{2^n}\right)-\left(f(x)+\tan\frac{x}{2^n}\right)^n \cdot\left[\sin\left(\tan\frac{x}{2}\right)\right]}{1+\left(f(x)+\tan\frac{x}{2^n}\right)^n}$ for $x \neq \dfrac{\pi}{4}$

= k for $x = \dfrac{\pi}{4}$ and the domain of g (x) is (0 , π/2).

where [ ] denotes the greatest integer function.

Find the value of k, if possible, so that g (x) is continuous at x = π/4. Also state the points of discontinuity of g (x) in (0 , π/4) , if any.

9. Let f (x) = $x^3 - x^2 - 3x - 1$ and h (x) = $\frac{f(x)}{g(x)}$ where *h* is a rational function such that

(a) it is continuous every where except when x = – 1, (b) $\lim_{x\to\infty} h(x) = \infty$ and

(c) $\lim_{x\to -1} h(x) = \frac{1}{2}$.

Find $\lim_{x\to 0} \left(3h(x) + f(x) - 2g(x)\right)$

10. Let f be continuous on the interval [0, 1] to R such that f (0) = f (1). Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that f (c) = f $\left(c + \frac{1}{2}\right)$

11. Consider the function g(x) = $\begin{cases} \dfrac{1 - a^x + x a^x \ell n a}{a^x x^2} & \text{for } x < 0 \\ \dfrac{2^x a^x - x\ell n 2 - x \ell n a - 1}{x^2} & \text{for } x > 0 \end{cases}$ where a > 0.

find the value of 'a' & 'g(0)' so that the function g(x) is continuous at x = 0.

12. Let f(x) = $\begin{cases} \dfrac{\left(\frac{\pi}{2} - \sin^{-1}\left(1 - \{x\}^2\right)\right)\sin^{-1}\left(1 - \{x\}\right)}{\sqrt{2}\left(\{x\} - \{x\}^3\right)} & \text{for } x \neq 0 \\ \dfrac{\pi}{2} & \text{for } x = 0 \end{cases}$ where {x} is the fractional part of x.

Consider another function g(x) ; such that

g(x) = f(x) for $x \geq 0$

= $2\sqrt{2}$ f(x) for x < 0

Discuss the continuity of the functions f(x) & g(x) at x = 0.

13. Discuss the continuity of f in [0,2] where f(x) = $\begin{cases} |4x - 5|\,[x] & \text{for } x > 1 \\ [\cos \pi x] & \text{for } x \leq 1 \end{cases}$; where [x] is the greatest integer not greater than x. Also draw the graph.

14. Let g(x) = $\tan^{-1}|x| - \cot^{-1}|x|$, f(x) = $\frac{[x]}{[x+1]}$ {x}, h(x) = $|g(f(x))|$ where {x} denotes fractional part and [x] denotes the integral part then which of the following holds good?

(A) h is continuous at x = 0 (B) h is discontinuous at x = 0

(C) $h(0^-) = \pi/2$ (D) $h(0^+) = -\pi/2$

15. f (x) = $\begin{cases} \sin\left(\frac{a-x}{2}\right)\tan\left[\frac{\pi x}{2a}\right] & \text{for } x > a \\ \dfrac{\left[\cos\left(\frac{\pi x}{2a}\right)\right]}{a - x} & \text{for } x < a \end{cases}$

where [x] is the greatest integer function of x, and a > 0, then

(A) $f(a^-) < 0$ (B) f has a removable discontinuity at x = a

(C) f has an irremovable discontinuity at x=a (D) $f(a^+) < 0$

EXERCISE-III

1. *f* is a continuous function on the real line. Given that for all real values of x,

$x^2 + (f(x) - 2)\,x - \sqrt{3}\cdot f(x) + 2\sqrt{3} - 3 = 0$. Then the value of $f(\sqrt{3})$

(A) can not be determined (B) is $2(1-\sqrt{3})$ (C) is zero (D) is $\dfrac{2\left(\sqrt{3}-2\right)}{\sqrt{3}}$

2. If $f(x) = \text{sgn}(\cos 2x - 2\sin x + 3)$, where sgn ( ) is the signum function, then f (x)

(A) is continuous over its domain (B) has a missing point discontinuity

(C) has isolated point discontinuity (D) has irremovable discontinuity.

3. Consider $f(x) = \underset{n\to\infty}{\text{Limit}} \dfrac{x^n - \sin x^n}{x^n + \sin x^n}$ for $x > 0, x \neq 1$,

$f(1) = 0$

then

(A) f is continuous at x = 1 (B) f has an infinite or oscillatory discontinuity at x = 1.

(C) f has a finite discontinuity at x = 1 (D) f has a removable type of discontinuity at x = 1.

4. Given $f(x) = \dfrac{[\{|x|\}]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)\text{sgn}(\sin x)}$ for $x \neq 0$

$= 0$ for $x = 0$

where {x} is the fractional part function; [x] is the step up function and sgn(x) is the signum function of x then, f(x)

(A) is continuous at x = 0 (B) is discontinuous at x = 0

(C) has a removable discontinuity at x = 0 (D) has an irremovable discontinuity at x = 0

5. Consider $f(x) = \begin{cases} x[x]^2 \log_{(1+x)} 2 & \text{for} -1 < x < 0 \\ \dfrac{\ln\left(e^{x^2} + 2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & \text{for} 0 < x < 1 \end{cases}$

where [*] & {*} are the greatest integer function & fractional part function respectively, then

(A) $f(0) = \ln 2 \Rightarrow$ f is continuous at x = 0 (B) $f(0) = 2 \Rightarrow$ f is continuous at x = 0

(C) $f(0) = e^2 \Rightarrow$ f is continuous at x = 0 (D) f has an irremovable discontinuity at x = 0

6. Consider $f(x) = \frac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}}$, $x \neq 0$; $g(x) = \cos 2x$, $-\frac{\pi}{4} < x < 0$,

$$h(x) - \begin{cases} \frac{1}{\sqrt{2}} f(g(x)) & \text{for } x<0 \\ 1 & \text{for } x=0 \\ f(x) & \text{for } x>0 \end{cases}$$

then, which of the following holds good.
where {x} denotes fractional part function.

(A) 'h' is continuous at x = 0
(B) 'h' is discontinuous at x = 0
(C) f(g(x)) is an even function
(D) f(x) is an even function

7. The function $f(x) = [x].\cos\frac{2x-1}{2}\pi$, where [•] denotes the greatest integer function, is discontinuous at

(A) all x
(B) all integer points
(C) no x
(D) x which is not an integer

8. Consider the function defined on [0, 1] $\to$ R, $f(x) = \frac{\sin x - x\cos x}{x^2}$ if $x \neq 0$ and f (0) = 0, then the function $f(x)$

(A) has a removable discontinuity at x = 0
(B) has a non removable finite discontinuity at x = 0
(C) has a non removable infinite discontinuity at x = 0
(D) is continuous at x = 0

9. Consider the function $f(x) = \lim_{n\to\infty} \frac{\sin \pi x - x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}}$, where $n \in N$

Statement-1: $f(x)$ is discontinuous at x = 1.
because
Statement-2: $f(1) = 0$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

10. Consider the functions, f (x) = sgn (x – 1) and g (x) = $\cot^{-1}[x-1]$
where [ ] denotes the greatest integer function.

Statement-1 : The function F (x) = f (x) · g (x) is discontinuous at x = 1.
because
Statement-2 : If f (x) is discontinuous at x = a and g (x) is also discontinuous at x = a then the product function f (x) · g (x) is discontinuous at x = a.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

11. $f(x) = \dfrac{a\cos x - \cos bx}{x^2}$, when $x \neq 0$ and $f(0) = 4$. If f(x) is continuous at x = 0, then the value of $(n + \Sigma a_i + \Sigma b_i)$ is

[Here, n = number of ordered pairs (a, b);
Σa_i = sum of all distinct possible values of a ;
Σb_i = sum of all distinct possible values of b]

(A) 2 (B) 3 (C) 4 (D) 5

12. **Statement-1 :** Function f(x) = [cos x] is discontinuous at $x = \pi$.

Statement-2 : Function g(x) = [x] is discontinuous at integral values of x.

[Here, [·] denotes greatest integer function, i.e., [k] equals the greatest integer less than or equal to 'k']

(A) Statement-1 is true, Statement-2 is true & Statement-2 is correct explanation for Statement-1

(B) Statement-1 is true, Statement-2 is true & Statement-2 is **NOT** the correct explanation for Statement-1

(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is false, Statement-2 is true

13. Consider the function $f(x) = \begin{cases} x^2 + ax + 2, & \text{if } x \text{ is rational} \\ 1 - x, & \text{if } x \text{ is irrational} \end{cases}$, where 'a' is a parameter that can take any real value. If f(x) is discontinuous for all real values of x, then number of possible integral values of a is

(A) 3 (B) 2 (C) 1 (D) 0

14. If $f(x) = [x] + \left[x + \dfrac{1}{3}\right] + \left[x + \dfrac{2}{3}\right]$, then number of points of discontinuity of f(x) in [–1, 1] is :

(A) 5 (B) 4 (C) 7 (D) 6

15. If $f(x) = \text{sgn}(\sin^2 x - \sin x - 1)$ has exactly four points of discontinuity for $x \in (0, n\pi)$, $n \in N$, then

(A) the minimum possible value of n is 5

(B) the maximum possible value of n is 6

(C) there are exactly two possible values of n

(D) None of these

EXERCISE-IV

1. Let $f(x) = \frac{1-\tan x}{4x-\pi}$, $x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$. If f(x) is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is- **[AIEEE 2004]**

(A) 1 (B) 1/2 (C) – 1/2 (D) –1

2. The function $f : R - \{0\} \to R$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$ can be made continuous at x = 0 by defining *f*(0) as

(A) 2 (B) –1 (C) 0 (D) 1 **[AIEEE 2007]**

3. The value of p and q for which the function **[AIEEE 2011]**

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & x < 0 \\ q \ , & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} & x > 0 \end{cases}$$ is continuous for all $x \in R$, are :

(A) $p = \frac{1}{2}, q = \frac{3}{2}$ (B) $p = \frac{1}{2}, q = -\frac{3}{2}$ (C) $p = \frac{5}{2}, q = \frac{1}{2}$ (D) $p = -\frac{3}{2}, q = \frac{1}{2}$

4. If $f : R \to R$ is a function defined by $f(x) = [x] \cos\left(\frac{2x-1}{2}\right)\pi$, where [x] denotes the greatest integer function, then f is : **[AIEEE 2012]**

(A) continuous for every real x. (B) discontinuous only at x = 0

(C) discontinuous only at non-zero integral values of x. (D) continuous only at x = 0

5. Let $f(x) = \begin{cases} (x-1)^{\frac{1}{2-x}}, & x > 1, x \neq 2 \\ k & , x = 2 \end{cases}$ **[IIT Main 2018]**

The value of k for which f is continuus at x = 2 is :

(A) 1 (B) e (C) e^{-1} (D) e^{-2}

6. If the function f defined as $f(x) = \frac{1}{x} - \frac{k-1}{e^{2x}-1}$, $x \neq 0$, is continuous at x = 0, then the ordered pair (k, f(0)) is equal to **[IIT Main Online 2018]**

(A) (2, 1) (B) (3, 1) (C) $\left(\frac{1}{3}, 2\right)$ (D) (3, 2)

EXERCISE-V

1. The function $f(x) = [x]^2 - [x^2]$ (where [y] is the greatest integer less than or equal to y), is discontinuous at :
(A) all integers (B) all integers except 0 & 1 **[JEE '99, 2 (out of 200)]**
(C) all integers except 0 (D) all integers except 1

2. Determine the constants a, b & c for which the function $f(x) = \begin{cases} (1+ax)^{1/x} & \text{for } x < 0 \\ b & \text{for } x = 0 \\ \frac{(x+c)^{1/3} - 1}{(x+1)^{1/2} - 1} & \text{for } x > 0 \end{cases}$ is continuous at x = 0.

[REE '99, 6]

3. Discuss the continuity of the function

$$f(x) = \begin{cases} \frac{e^{1/(x-1)} - 2}{e^{1/(x-1)} + 2}, & x \neq 1 \\ 1, & x = 1 \end{cases}$$

at x = 1. **[REE 2001 (Mains) , 3 out of 100]**

4. For every integer n, let a_n and b_n be real numbers. Let function $f : R \to R$ be given by **[JEE 2012]**

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}, \text{ for all integers n.}$$

If f is continuous , then which of the following hold(s) for all n ?
(A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

5. For every pair of continuous functions $f, g : [0, 1] \to R$, such that $\max \{f(x) : x \in [0, 1]\} = \max\{g(x): x \in [0, 1]\}$, the correct statement(s) is(are) : **[IIT Advance 2014]**
(A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
(B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
(C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
(D) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

6. Let [x] be the greatest integer less than or equals to x. Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous? **[IIT Advance - 2017]**
(A) x = 1 (B) x = –1 (C) x = 0 (D) x = 2

ANSWER KEY

PROFICIENCY TEST

1. – 1 **2.** a = 0, b = 1 **3.** b = 1 and a, c ∈ R such that a + c = 3

4. f is cont. in $-1 \le x \le 1$ **5.** P not possible. **6.** (a) $4 - 3\sqrt{2} + a$, (b) a = 3

7. g(x) = 2 + x for $0 \le x \le 1$, $2 - x$ for $1 < x \le 2$, $4 - x$ for $2 < x \le 3$, g is discontinuous at x = 1 & x = 2

8. A = 1 ; f(2) = 1/2 **9.** discontinuous at x = 1, 4 & 5

10. **(i)** false ; **(ii)** false ; **(iii)** true ; **(iv)** true; **(v)** true; **(vi)** true

EXERCISE-I

1. $f(0^+) = -2$; $f(0^-) = 2$ hence f(0) not possible to define

2. $y_n(x)$ is continuous at x = 0 for all n and y(x) is dicontinuous at x = 0

3. a = 0 ; b = –1 **4.** a = 0, b = 1 **5.** gof is dis-continuous at x = 0, 1 & – 1

6. a = 1/2, b = 4 **7.** $a = -3/2$, $b \neq 0$, c = 1/2 **8.** $A = -4$, B = 5, f(0) = 1

9. A, C, D **10** A, C, D **11.** B, C **12.** C, D

13. B **14.** B, C, D **15.** B, C

EXERCISE-II

1. discontinuous at all integral values in $[-2, 2]$

2. locus (a, b) $\rightarrow$ x, y is y = x – 3 excluding the points where y = 3 intersects it.

3. 5 **5.** $\frac{1}{60}$

8. k = 0 ; $g(x) = \begin{cases} \ell n\,(\tan x) & \text{if } 0 < x < \frac{\pi}{4} \\ 0 & \text{if } \frac{\pi}{4} \le x < \frac{\pi}{2} \end{cases}$. Hence g (x) is continuous everywhere.

9. g (x) = 4 (x + 1) and limit = $-\frac{39}{4}$ **11.** $a = \frac{1}{\sqrt{2}}$, $g(0) = \frac{(\ell n 2)^2}{8}$

12. $f(0^+) = \frac{\pi}{2}$; $f(0^-) = \frac{\pi}{4\sqrt{2}}$ $\Rightarrow$ f is discontinuous at x = 0 ;

$g(0^+) = g(0^-) = g(0) = \pi/2 \Rightarrow$ g is continuous at x = 0

13. the function f is continuous everywhere in [0 , 2] except for x = 0 , $\frac{1}{2}$, 1 & 2.

14. A **15.** B

EXERCISE-III

1. B **2.** C **3.** C **4.** A **5.** D **6.** A **7.** C

8. D **9.** B **10.** C **11.** B **12.** D **13.** A **14.** D

15. C

EXERCISE-IV

1. C **2.** D **3.** D **4.** A **5.** C **6.** B

EXERCISE-V

1. D **2.** $a = \ell n\frac{2}{3}$; $b = \frac{2}{3}$; c = 1

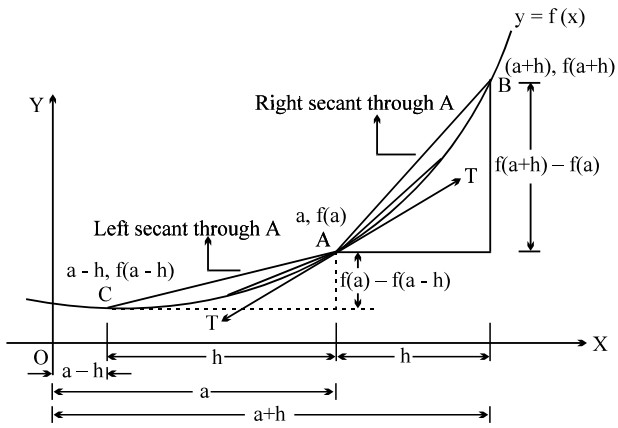
3. Discontinuous at x = 1; $f(1^+) = 1$ and $f(1^-) = -1$ **4.** BD **5.** AD **6.** A,B,D

DIFFERENTIABILITY
KEY CONCEPTS

THINGS TO REMEMBER :

1. Right hand & Left hand Derivatives ;

By definition : $f'(a) = \underset{x\to a}{\text{Limit}} \frac{f(x)-f(a)}{x-a}$ if it exist



(i) The right hand derivative of f ′ at x = a

denoted by $f'(a^+)$ is defined by :

$$f'(a^+) = \underset{h\to 0^+}{\text{Limit}} \frac{f(a+h)-f(a)}{h},$$

provided the limit exists & is finite.

(ii) The left hand derivative : of f at x = a

denoted by $f'(a^-)$ is defined by :

$$f'(a^-) = \underset{h\to 0^+}{\text{Limit}} \frac{f(a-h)-f(a)}{-h},$$

Provided the limit exists & is finite.

We also write $f'(a^+) = f'_+(a)$ & $f'(a^-) = f'_-(a)$.

* This geomtrically means that a unique tangent with finite slope can be drawn at x = a as shown in the figure.

(iii) Derivability & Continuity :

(a) If f ′(a) exists then f(x) is derivable at x= a $\Rightarrow$ f(x) is continuous at x = a.

(b) If a function f is derivable at x then f is continuous at x.

For : $f'(x) = \underset{h\to 0}{\text{Limit}} \frac{f(x+h)-f(x)}{h}$ exists.

Also $f(x+h)-f(x) = \frac{f(x+h)-f(x)}{h}.h \; [h \neq 0]$

Therefore :

$$\underset{h\to 0}{\text{Limit}} [f(x+h)-f(x)] = \underset{h\to 0}{\text{Limit}} \frac{f(x+h)-f(x)}{h}.h = f'(x).0 = 0$$

Therefore $\underset{h\to 0}{\text{Limit}} [f(x+h)-f(x)] = 0 \Rightarrow \underset{h\to 0}{\text{Limit}} f(x+h) = f(x) \Rightarrow$ f is continuous at x.

Note : If f(x) is derivable for every point of its domain of definition, then it is continuous in that domain.
The Converse of the above result is not true :

" IF f IS CONTINUOUS AT x , THEN f IS DERIVABLE AT x " IS NOT TRUE.

e.g. the functions $f(x) = |x|$ & $g(x) = x \sin\frac{1}{x}$; $x \neq 0$ & $g(0) = 0$ are continuous at x = 0 but not derivable at x = 0.

NOTE CAREFULLY :

(a) Let $f'_+(a) = p$ & $f'_-(a) = q$ where p & q are finite then :

(i) $p = q \Rightarrow$ f is derivable at $x = a \Rightarrow$ f is continuous at x = a.

(ii) $p \neq q \Rightarrow$ f is not derivable at x = a. $\Rightarrow$ f is continuous at x = a.

In short, for a function f :

Differentiability $\Rightarrow$ Continuity ; Continuity $\not\Rightarrow$ derivability ;

Non derivibality $\not\Rightarrow$ discontinuous; But discontinuity $\Rightarrow$ Non derivability

(b) If a function f is not differentiable but is continuous at x = a it geometrically implies a sharp corner at x = a.

3. DERIVABILITY OVER AN INTERVAL :

f (x) is said to be derivable over an interval if it is derivable at each & every point of the interval f(x) is said to be derivable over the closed interval [a, b] if :

(i) for the points a and b, $f'(a+)$ & $f'(b-)$ exist &

(ii) for any point c such that $a < c < b$, $f'(c+)$ & $f'(c-)$ exist & are equal.

NOTE :

1. If f(x) & g(x) are derivable at x = a then the functions f(x) + g(x), f(x) – g(x) , f(x).g(x) will also be derivable at x = a & if $g(a) \neq 0$ then the function f(x)/g(x) will also be derivable at x = a.

2. If f(x) is differentiable at x = a & g(x) is not differentiable at x = a , then the product function $F(x) = f(x) \cdot g(x)$ can still be differentiable at x = a e.g. $f(x) = x$ & $g(x) = |x|$.

3. If f(x) & g(x) both are not differentiable at x = a then the product function ;
$F(x) = f(x) \cdot g(x)$ can still be differentiable at x = a e.g. $f(x) = |x|$ & $g(x) = |x|$.

4. If f(x) & g(x) both are non-deri. at x = a then the sum function F(x) = f(x) + g(x) may be a differentiable function.
e.g. $f(x) = |x|$ & $g(x) = -|x|$.

5. If f(x) is derivable at x = a $\not\Rightarrow$ $f'(x)$ is continuous at x = a.

e.g. $f(x) = \begin{cases} x^2 \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

PROFICIENCY TEST

1. Discuss the continuity & differentiability of the function f(x) = sinx + sin $|x|$, $x \in R$. Draw a rough sketch of the graph of f(x).

2. Examine the continuity and differentiability of f(x) = $|x| + |x-1| + |x-2|$, $x \in R$.
Also draw the graph of f(x).

3. A function *f* is defined as follows : f(x) = $\begin{cases} 1 & \text{for} \quad -\infty<x<0 \\ 1+|\sin x| & \text{for} \quad 0\le x<\frac{\pi}{2} \\ 2+\left(x-\frac{\pi}{2}\right)^2 & \text{for} \quad \frac{\pi}{2}\le x<+\infty \end{cases}$

Discuss the continuity & differentiability at x = 0 & x = $\pi/2$.

4. Examine the origin for continuity & derrivability in the case of the function f defined by
$f(x) = x \tan^{-1}(1/x)$, $x \neq 0$ and f(0) = 0.

5. Discuss the continuity & the derivability of 'f' where f (x) = degree of $(u^{x^2} + u^2 + 2u - 3)$ at x = $\sqrt{2}$

Fill in the blanks :

6. If f(x) is derivable at x = 3 & $f'(3) = 2$, then $\underset{h\to 0}{\text{Limit}} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ = _____.

7. If f(x) = $|\sin x|$ & $g(x) = x^3$ then f[g(x)] is _____ & _____ at x = 0. (State continuity and derivability)

8. Let f(x) be a function satisfying the condition $f(-x) = f(x)$ for all real x. If $f'(0)$ exists, then its value is _____.

9. For the function f(x) = $\begin{cases} \frac{x}{1+e^{1/x}}, x\neq 0 \\ 0 \quad , x=0 \end{cases}$, the derivative from the right, $f'(0^+)$ = _____ & the derivative from the left, $f'(0^-)$ = _____.

10. The number of points at which the function f(x) = max. $\{a-x, a+x, b\}$, $-\infty < x < \infty$, $0 < a < b$ cannot be differentiable is _____.

EXERCISE-I

1. Given a differentiable function $f(x)$ defined for all real x, and is such that $f(x+h) - f(x) \le 6h^2$ for all real h and x. Show that $f(x)$ is constant.

2. Let $f(0) = 0$ and $f'(0) = 1$. For a positive integer k, show that

$$\lim_{x\to 0} \frac{1}{x}\left(f(x) + f\left(\frac{x}{2}\right) + ...f\left(\frac{x}{k}\right)\right) = 1 + \frac{1}{2} + \frac{1}{3} + ...+ \frac{1}{k}$$

3. Let $f(x) = xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}$, $x \neq 0$; $f(0) = 0$. Test the continuity & differentiability at x = 0.

4. If $f(x) = |x-1| . ([x] - [-x])$, then find $f'(1^+)$ & $f'(1^-)$ where [x] denotes greatest integer function.

5. If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \ge 1 \end{cases}$ is derivable at x = 1. Find the values of a & b.

6. Consider the functions $f(x) = x^2 - 2x$ and $g(x) = -|x|$

Statement-1: The composite function $F(x) = f(g(x))$ is not derivable at x = 0.

because

Statement-2: $F'(0^+) = 2$ and $F'(0^-) = -2$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Select the correct alternative : (More than one are correct)

7. $f(x) = |x[x]|$ in $-1 \le x \le 2$, where [x] is greatest integer $\le x$ then f(x) is :

(A) continuous at x = 0 (B) discontinuous x = 0

(C) not differentiable at x = 2 (D) differentiable at x = 2

8. $f(x) = 1 + x.[\cos x]$ in $0 < x \le \pi/2$, where [ ] denotes greatest integer function then ,

(A) It is continuous in $0 < x < \pi/2$ (B) It is differentiable in $0 < x < \pi/2$

(C) Its maximum value is 2 (D) It is not differentiable in $0 < x < \pi/2$

9. $f(x) = (\sin^{-1}x)^2 . \cos(1/x)$ if $x \neq 0$; $f(0) = 0$, f(x) is :

(A) continuous no where in $-1 \le x \le 1$ (B) continuous every where in $-1 \le x \le 1$

(C) differentiable no where in $-1 \le x \le 1$ (D) differentiable everywhere in $-1 < x < 1$

10. $f(x) = |x| + |\sin x|$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. It is :

(A) Continuous no where (B) Continuous every where

(C) Differentiable no where (D) Differentiable everywhere except at x = 0

11. If $f(x) = 2 + |\sin^{-1} x|$, it is :

(A) continuous no where (B) continuous everywhere in its domain

(C) differentiable no where in its domain (D) Not differentiable at x = 0

12. If $f(x) = x^2 . \sin(1/x)$, $x \neq 0$ and $f(0) = 0$ then ,

(A) f(x) is continuous at x = 0 (B) f(x) is derivable at x = 0

(C) f′(x) is continuous at x = 0 (D) f′′(x) is not derivable at x = 0

13. A function which is continuous & not differentiable at x = 0 is :

(A) $f(x) = x$ for $x < 0$ & $f(x) = x^2$ for $x \geq 0$ (B) $g(x) = x$ for $x < 0$ & $g(x) = 2x$ for $x \geq 0$

(C) $h(x) = x|x|$ $x \in R$ (D) $K(x) = 1 + |x|$, $x \in R$

14. If $\sin^{-1}x + |y| = 2y$ then y as a function of x is :

(A) defined for $-1 \leq x \leq 1$ (B) continuous at x = 0

(C) differentiable for all x (D) such that $\frac{dy}{dx} = \frac{1}{3\sqrt{1-x^2}}$ for $-1 < x < 0$

15. Let $f(x) = \cos(x)$ & $H(x) = \begin{cases} \text{Min}[f(t)/0 \leq t \leq x] & \text{for } 0 \leq x \leq \frac{\pi}{2} \\ \frac{\pi}{2} - x & \text{for } \frac{\pi}{2} < x \leq 3 \end{cases}$, then

(A) H (x) is continuous & derivable in [0, 3]

(B) H(x) is continuous but not derivable at $x = \pi/2$

(C) H(x) is neither continuous nor derivable at $x = \pi/2$

(D) Maximum value of H(x) in [0,3] is 1

EXERCISE-II

1. Let f(x) be defined in the interval [-2, 2] such that $f(x) = \begin{cases} -1 & , -2 \leq x \leq 0 \\ x - 1 & , 0 < x \leq 2 \end{cases}$ &

$g(x) = f(|x|) + |f(x)|$. Test the differentiability of g(x) in $(-2, 2)$.

2. Given $f(x) = \cos^{-1}\left(\text{sgn}\left(\frac{2[x]}{3x - [x]}\right)\right)$ where sgn (.) denotes the signum function & [.] denotes the greatest integer function. Discuss the continuity & differentiability of f (x) at $x = \pm 1$.

3. Examine for continuity & differentiability the points x = 1 & x = 2, the function f defined by $f(x) = \begin{cases} x[x] & , 0 \leq x < 2 \\ (x-1)[x] & , 2 \leq x \leq 3 \end{cases}$ where [x] = greatest integer less than or equal to x.

4. $f(x) = x\left(\frac{e^{[x]+|x|}-2}{[x]+|x|}\right)$, $x \neq 0$ & $f(0) = -1$ where [x] denotes greatest integer less than or equal to x. Test the differentiability of f(x) at x = 0.

5. Discuss the continuity & the derivability in [0 , 2] of $f(x) = \begin{cases} |2x-3|[x] & \text{for } x \geq 1 \\ \sin\frac{\pi x}{2} & \text{for } x < 1 \end{cases}$

where [ ] denote greatest integer function .

6. If $f(x) = -1 + |x-1|$, $-1 \le x \le 3$; $g(x) = 2 - |x+1|$, $-2 \le x \le 2$, then calculate (fog) (x) & (gof) (x). Draw their graph. Discuss the continuity of (fog) (x) at x = –1 & the differentiability of (gof) (x) at x = 1.

7. The function $f(x) = \begin{cases} ax(x-1)+b & \text{when } x<1 \\ x-1 & \text{when } 1 \le x \le 3 \\ px^2+qx+2 & \text{when } x>3 \end{cases}$

Find the values of the constants a, b, p, q so that
(i) f(x) is continuous for all x (ii) f ' (1) does not exist (iii) f '(x) is continuous at x = 3

8. Examine the function, $f(x) = x.\dfrac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ ($a > 0$) and f (0) = 0 for continuity and existence of the derivative at the origin.

9. $f(x) = \begin{cases} 1-x & , \ (0 \le x \le 1) \\ x+2 & , \ (1 < x < 2) \\ 4-x & , \ (2 \le x \le 4) \end{cases}$ Discuss the continuity & differentiability of y = f [f(x)] for $0 \le x \le 4$.

10. f : R $\to$ R be a function satisfying $f(x + 2y) = f(x) + f(2y) + 4xy + e^x \cdot e^{2y} - e^x - e^{2y}$ $\forall$ x, y $\in$ R. If $f'(0) = 1$, find f(x) in terms of x.

11. Let *f* (x) be a function defined on (–a, a) with a > 0. Assume that *f* (x) is continuous at x = 0 and $\lim\limits_{x \to 0} \dfrac{f(x) - f(kx)}{x} = \alpha$, where k $\in$ (0, 1) then compute f ' (0^+) and f ' (0^-), and comment upon the differentiability of *f* at x = 0.

12. Consider the function, $f(x) = \begin{cases} x^2 \left|\cos\dfrac{\pi}{2x}\right| & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

(a) Show that *f* ' (0) exists and find its value (b) Show that *f* ' $(1/3)$ does not exist.

(c) For what values of x, *f* ' (x) fails to exist.

13. Let f(x) be a real valued function not identically zero satisfies the equation, $f(x + y^n) = f(x) + (f(y))^n$ for all real x & y and $f'(0) \ge 0$ where n (> 1) is an odd natural number. Find f(10).

14. A derivable function $f: R^+ \to R$ satisfies the condition $f(x) - f(y) \ge \ln \dfrac{x}{y} + x - y$ for every x, y $\in R^+$. If g denotes the derivative of *f* then compute the value of the sum $\sum\limits_{n=1}^{100} g\left(\dfrac{1}{n}\right)$.

15. Suppose that f & g are non constant differentiable, real valued functions on R. If for every x, y $\in$ R, $f(x + y) = f(x) f(y) - g(x) g(y)$; $g(x + y) = g(x) f(y) + f(x) g(y)$ & $f'(0) = 0$ then prove that $f^2(x) + g^2(x) = 1$, $\forall$ x $\in$ R.

EXERCISE-III

1. Let $f(x) = \dfrac{|x|}{\sin x}$ for $x \neq 0$ & $f(0) = 1$ then ,

(A) f(x) is continuous & derivable at x = 0
(B) f(x) is continuous & not derivable at x = 0
(C) f(x) is discontinuous at x = 0
(D) none

2. Given $f(x) = \begin{cases} \log_a \left(a^{|[x] + [-x]|} \right)^x \left(\dfrac{a^{\frac{2}{\left(\frac{[x] + [-x]}{|x|}\right)} - 5}}{3 + a^{\frac{1}{|x|}}} \right) & \text{for } |x| \neq 0 \ ; \ a > 1 \\ 0 & \text{for } x = 0 \end{cases}$ where [ ] represents the integral part function, then :

(A) f is continuous but not differentiable at x = 0
(B) f is cont. & diff. at x = 0
(C) the differentiability of 'f' at x = 0 depends on the value of a
(D) f is cont. & diff. at x = 0 and for a = e only.

3. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases}$ is differentiable for all real x ?

(A) $\{(a, 1-2a, a) \mid a \in R, a \neq 0\}$
(B) $\{(a, 1-2a, c) \mid a, c \in R, a \neq 0\}$
(C) $\{(a, b, c) \mid a, b, c \in R, a + b + c = 1\}$
(D) $\{(a, 1-2a, 0) \mid a \in R, a \neq 0\}$

4. A function f defined as $f(x) = x[x]$ for $-1 \leq x \leq 3$ where [x] defines the greatest integer $\leq x$ is :
(A) continuous at all points in the domain of f but non-derivable at a finite number of points
(B) discontinuous at all points & hence non-derivable at all points in the domain of f (x)
(C) discontinuous at a finite number of points but not derivable at all points in the domain of f (x)
(D) discontinuous & also non-derivable at a finite number of points of f (x).

5. [x] denotes the greatest integer less than or equal to x. If $f(x) = [x] [\sin \pi x]$ in $(-1,1)$ then f(x) is :
(A) continuous at x = 0
(B) continuous in $(-1, 0)$
(C) differentiable in $(-1,1)$
(D) none

6. The function f(x) is defined as follows $f(x) = \begin{cases} -x & \text{if} \quad x<0 \\ x^2 & \text{if} \quad 0 \leq x \leq 1 \\ x^3 - x + 1 & \text{if} \quad x>1 \end{cases}$ then f(x) is :

(A) derivable and continuous at x = 0
(B) derivable at x = 1 but not continuous at x = 1
(C) neither derivable nor continuous at x = 1
(D) not derivable at x = 0 but continuous at x = 1

7. If $f(x) = \begin{cases} x + \{x\} + x\sin\{x\} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ where {x} denotes the fractional part function, then :

(A) 'f' is continuous & diff. at x = 0 (B) 'f' is continuous but not diff. at x = 0
(C) 'f' is continuous & diff. at x = 2 (D) none of these

8. The set of all points where the function $f(x) = \frac{x}{1+|x|}$ is differentiable is :

(A) $(-\infty, \infty)$ (B) $[0, \infty)$ (C) $(-\infty, 0) \cup (0, \infty)$ (D) $(0, \infty)$

9. Let f be an injective and differentiable function such that $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy)$ for all non negative real x and y with $f'(0) = 0$, $f'(1) = 2 \neq f(0)$, then
(A) $x f'(x) - 2 f(x) + 2 = 0$ (B) $x f'(x) + 2 f(x) - 2 = 0$
(C) $x f'(x) - f(x) + 1 = 0$ (D) $2 f(x) = f'(x) + 2$

10. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ and p is a prime number. The number of points where $f(x)$ is not differentiable is
(A) p – 1 (B) p + 1 (C) 2p + 1 (D) 2p – 1.
Here [x] denotes greatest integer function.

11. The function $f(x) = \sin^{-1}(\cos x)$ is :

(A) discontinuous at $x = 2\pi$ (B) differentiable at $x = \frac{3\pi}{2}$

(C) not differentiable at $x = \frac{\pi}{2}$ (D) differentiable at $x = 2\pi$

12. $f_1(x) = (x^2 - 4)\,|(x-2)(x-3)|$; $f_2(x) = \sin(|x-2|) - |x-2|$, $f_3(x) = \tan(|x-2|) + |x-2|$
How many of the above functions $\{f_1(x), f_2(x), f_3(x)\}$ are differentiable at x = 2 ?
(A) 0 (B) 1 (C) 2 (D) 3

13. Consider the function $f(x) = \max\{||x-2|-1|, \alpha\}$, where, α is some real number. Find the number of values of $x \in \mathbf{R}$ at which f(x) is non-differentiable if $\alpha = \frac{1}{2}$

(A) 2 (B) 3 (C) 4 (D) 5

14. The function $f(x) = \frac{|x| - x(3^{1/x} + 1)}{3^{1/x} - 1}$, $x \neq 0$, $f(0) = 0$ is :

(A) discontinuous at x = 0 (B) continuous at x = 0 but not differentiable there
(C) both continuous and differentiable at x = 0 (D) differentiable but not continuous at x = 0

15. If $f(x) = |1 - x|$, then the points where $\sin^{-1}(f|x|)$ is non-differentiable are :
(A) {0, 1} (B) {0, –1} (C) {0, 1, –1} (D) None of these

EXERCISE-IV

1. If $f(x) = \begin{cases} x e^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x \neq 0 \\ 0 \quad\quad\quad , & x = 0 \end{cases}$ then f(x) is **[AIEEE 2003]**

(A) discontinuous everywhere (B) continuous as well as differentiable for all x

(C) continuous for all x but not differentiable at x = 0 (D) neither differentiable nor continuous at x = 0

2. If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x-y)^2$, x, y $\in$ R and f(0) = 0, then f(1) equals **[AIEEE-2005]**

(A) –1 (B) 0 (C) 2 (D) 1

3. Suppose f(x) is differentiable at x = 1 and $\lim_{h\to 0} \frac{1}{h} f(1+h) = 5$, then f′(1) equals - **[AIEEE-2005]**

(A) 3 (B) 4 (C) 5 (D) 6

4. The set of points where $f(x) = \frac{x}{1+|x|}$ is differentiable is – **[AIEEE 2006]**

(A) $(-\infty, -1) \cup (-1, \infty)$ (B) $(-\infty, \infty)$ (C) $(0, \infty)$ (D) $(-\infty, 0) \cup (0, \infty)$

5. Let $f : R \to R$ be a function defined by $f(x) = \min \{x + 1, |x| + 1\}$. Then which of the following is true? **[AIEEE 2007]**

(A) $f(x) \geq 1$ for all $x \in R$ (B) $f(x)$ is not differentiable at x = 1

(C) $f(x)$ is differentiable everywhere (D) $f(x)$ is not differentiable at x = 0

6. Let $f(x) = \begin{cases} (x-1)\sin\frac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$

Then which one of the following is true ? **[AIEEE 2008]**

(A) f is differentiable at x = 0 and at x = 1 (B) f is differentiable at x = 0 but not at x = 1

(C) f is differentiable at x = 1 but not at x = 0 (D) f is neither differentiable at x = 0 nor at x = 1

7. If the function , $g(x) = \begin{cases} k\sqrt{x+1}, & 0 \leq x \leq 3 \\ mx+2, & 3 < x \leq 5 \end{cases}$

is differentiable, then the value of k + m is : **[IIT Mains - 2015]**

(A) 4 (B) 2 (C) $\frac{16}{5}$ (D) $\frac{10}{3}$

8. Let $S = \{ t \in R : f(x) = |x - \pi|. (e^{|x|} - 1) \sin|x| \text{ is not differentiable at } t\}$. Then the set S is equal to **[IIT Mains - 2018]**

(A) $\{0, \pi\}$ (B) ϕ (an empty set) (C) $\{0\}$ (D) $\{\pi\}$

EXERCISE-V

1. Let S = (0, 1) ∪ (1, 2) ∪ (3, 4) and T = {0,1, 2, 3}. Then which of the following statements is(are) true? **[JEE Advanced 2023]**

(A) There are infinitely many functions from S to T

(B) There are infinitely many strictly increasing functions from S to T

(C) The number of continuous functions from S to T is at most 120

(D) Every continuous function from S to T is differentiable

2. Let f : **R** → **R** and g : **R** → **R** be functions defined by

$$f(x) = \begin{cases} x\,|x|\sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x - 0, \end{cases} \text{ and } g(x) = \begin{cases} 1-2x, & 0 \le x \le \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Let a, b, c, d ∈ **R**. Define the function h : **R** → **R** by

$$h(x) = af(x) + b\left(g(x) + g\left(\frac{1}{2} - x\right)\right) + c(x - g(x)) + dg(x),\ x \in \mathbf{R}.$$

Match each entry in **List-I** to the correct entry in **List-II.**

	List-I		**List-II**
(P)	If a = 0, b = 1, c = 0, and d = 0, then	(1)	h is one-one
(Q)	If a = 1, b = 0, c = 0, and d = 0, then	(2)	h is onto
(R)	If a = 0, b = 0, c = 1, and d = 0, then	(3)	h is differentiable on **R**
(S)	If a = 0, b = 0, c = 0, and d = 1, then	(4)	the range of h is [0, 1]
		(5)	the range of h is {0, 1}

The correct option is : **[JEE Advanced 2024]**

(A) (P) → (4), (Q) → (3), (R) → (1), (S) → (2)

(B) (P) → (5), (Q) → (2), (R) → (4), (S) → (3)

(C) (P) → (5), (Q) → (3), (R) → (2), (S) → (4)

(D) (P) → (4), (Q) → (2), (R) → (1), (S) → (3)

3. Let f : ℝ → ℝ be a function defined by $f(x) = \begin{cases} x^2 \sin\left(\frac{\pi}{x^2}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$

Then which of the following statements is TRUE? **[JEE Advanced 2024]**

(A) f(x) = 0 has infinitely many solutions in the interval $\left[\frac{1}{10^{10}}, \infty\right)$.

(B) f(x) = 0 has no solutions in the interval $\left[\frac{1}{\pi}, \infty\right)$.

(C) The set of solutions of f(x) = 0 in the interval $\left(0, \frac{1}{10^{10}}\right)$ is finite.

(D) f(x) = 0 has more than 25 solutions in the interval $\left(\frac{1}{\pi^2}, \frac{1}{\pi}\right)$.

4. Let f : ℝ → ℝ be a function such that f(x + y) = f(x) + f(y) for all x, y ∈ ℝ, and g : ℝ → (0, ∞) be a function such that g(x + y) = g(x) g(y) for all x , y ∈ ℝ.

If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ is _____. **[JEE Advanced 2024]**

ANSWER KEY

PROFICIENCY TEST

1. f(x) is continuous but not derivable at x = 0 **2.** continuous $\forall$ $x \in R$, not derivable at x = 0,1 & 2

3. continuous but not derivable at x = 0 ; derivable & continuous at x = $\pi/2$

4. continuous but not derivable at x = 0 **5.** continuous but not derivable at x = $\sqrt{2}$

6. 2 **7.** continuous & derivable **8.** 0 **9.** $f'(0^+) = 0$, $f'(0^-) = 1$ **10.** 2

EXERCISE-I

3. f is continuous but not derivable at x = 0 **4.** $f'(1^+) = 3$, $f'(1^-) = -1$

5. a = 1/2 , b = 3/2 **6.** A **7.** A, C **8.** A, B **9.** B, D **10.** B, D

11. B, D **12.** A, B , D **13.** A, B, D **14.** A, B, D **15.** A, D

EXERCISE-II

1. not derivable at x = 0 & x = 1

2. f is continuous & derivable at $x = -1$ but f is neither continuous nor derivable at x = 1

3. discontinuous & not derivable at x = 1, continuous but not derivable at x = 2

4. not derivable at x = 0

5. f is continuous at x = 1 , 3/2 & discontinuous at x = 2, f is not derivable at x =1, 3/2 , 2

6. (fog)(x) = x+1 for $-2 \le x \le -1$, $-(x+1)$ for $-1 < x \le 0$ & $x - 1$ for $0 < x \le 2$.

(fog)(x) is continuous at x = –1, (gof)(x) = x+1 for $-1 \le x \le 1$ & $3 - x$ for $1 < x \le 3$.

(gof)(x) is not differentiable at x = 1

7. $a \neq 1, b = 0, p = \frac{1}{3}$ and $q = -1$

8. If $a \in (0, 1)$ $f'(0^+) = -1$; $f'(0^-) = 1 \Rightarrow$ continuous but not derivable

a = 1 ; f(x) = 0 which is constant $\Rightarrow$ continuous and derivable

If a > 1 $f'(0^-) = -1$; $f'(0^+) = 1 \Rightarrow$ continuous but not derivable

9. f is continuous but not derivable at x = 1, discontinuous at x = 2 & x = 3. continuous & derivable at all other points

10. $f(x) = x^2 + e^x \ \forall \ x \in R$ **11.** $f'(0) = \frac{\alpha}{1-k}$

12. **(a)** $f'(0) = 0$, **(b)** $f'\left(\frac{1^-}{3}\right) = -\frac{\pi}{2}$ and $f'\left(\frac{1^+}{3}\right) = \frac{\pi}{2}$, **(c)** $x = \frac{1}{2n+1}$ $n \in I$

13. $f(x) = x \Rightarrow f(10) = 10$ **14.** 5150

EXERCISE-III

1. C **2.** B **3.** A **4.** D **5.** B **6.** D **7.** D
8. A **9.** A **10.** D **11.** B **12.** C **13.** D **14.** B
15. C

EXERCISE-IV

1. C **2.** B **3.** C **4.** B **5.** C **6.** B **7.** B
8. B

EXERCISE–V

1. A,C,D **2.** C **3.** D **4.** 51.00

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